A Low-Ripple High-Frequency High-Voltage Power Supply for Ion Pumps

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***Abstract*—Ion pumps rely on high-voltage power supplies with low output ripple to generate ultra-high vacuum. This paper presents a 2 to 7 kV DC power supply with less than 10 Vpp voltage ripple designed for such applications. The circuit utilizes high-frequency inverters, planar PCB transformers, Dickson voltage multipliers, and ripple cancellation. The implemented prototype can maintain 0.1% ripple at light load while achieving 84% peak efficiency at 80 W.**

***Index Terms*—ion pump, ripple, Dickson multiplier, planar transformer**

I. Introduction

Regarded as one of the most effective method to generate ultra-high vacuum and widely used for scientific instruments like electron microscopes, the ion pump outperforms conventional turbo-molecular and primary pumps due to its enclosed and static nature [1], [2]. The key element of an ion pump is the Penning cell, which requires a 2 to 7 kV high voltage to ionize the gas and trap the electrons [2]. Therefore, commercially available ion pumps need to be powered by high-voltage DC-DC converters (often referred as ion pump controllers) that are capable of delivering maximum power from 40 to 400 W with a large voltage gain.

Because the stability of the applied electric field influences the performance of Penning cells, especially small output voltage ripple is required in ion pumps at light load when the pressure is close to vacuum [3], [4]. Conventional designs with switching frequencies less than a few hundreds of kilohertz solely rely on output capacitors to filter out the voltage ripple [3]–[5]. However, as high voltage rated capacitors inevitably involve large packages for insulation, the attempt to mitigate output ripples often results in a significant increase in the converter size. Although the multi-phase method proposed by Park et al. effectively reduces the ripple without increasing filtering capacitance, the added circuitry undermines such benefit when more than 2 phases are used [6]. In general, as the voltage ripple is inversely proportional to the product of frequency and filter volume, increasing the switching frequency in such converters should push further the Pareto frontier between the ripple and capacitor sizes. Hence, achieving a smaller ripple with the same filter size is possible with higher frequencies, and the exact ripple requirement can be maintained with a more compact converter. But it is worth noting that this frequency extension is not unlimited. Not only do components have self-resonant frequencies but the parasitic capacitance in rectifiers also degrades the ripple reduction. Therefore, a more accurate model is needed to design compact ripple-restricted multipliers at high frequency, which should include the parasitic capacitance and provide more realistic guidance for selecting bypassing capacitance.

The structure of a high voltage power converter often utilizes a transformer for voltage step up and galvanic isolation. Conventional designs tend to use transformers with large turn ratios since simple full bridge rectifiers are implemented and the entire voltage gain depends on the turn ratios [3], [7]. However, such transformers suffer from large parasitic capacitances and significant winding losses at higher frequencies due to proximity and skin effects, which can be challenging to implement at beyond 1 MHz. With voltage multipliers, it is possible to leverage more stages for higher voltage gain and thus reduce the required turn ratios of transformers [5]. Previous studies also demonstrate that it is possible to implement air-core print circuit board (PCB) transformers at high frequencies, which eliminates core losses and increases reproducibility with lower cost and weight.

In this paper, we propose a prototype of ion pump power supply operating at 1.35 MHz with air-core PCB planar transformers capable of outputting adjustable voltage from 2 kV to 7 kV. The maximum output power is 80 W with 84 % peak efficiency, and the output voltage ripple is approximately 0.1% when the load current is less than 1 mA.

II. A High-Frequency Parasitics Model of Dickson Voltage Multiplier

Compared to Cockcroft or Hybrid multipliers, Dickson multipliers tend to have better voltage regulation and less ripple, especially with more stages or significant load currents [8]. Hence, it is a more suitable circuit for ripple-restrictive applications such as ion pump power supplies. Fig. 1 displays the schematic of a m-stage Dickson multiplier, where V_s represents the peak input AC voltage with frequency f while I is the output DC current. Here we ignore the reverse recovery of diodes. Schottky diodes are preferred for high-frequency circuits and their reverse recovery loss is commonly negligible. We consider the identical idealized model for every diode with

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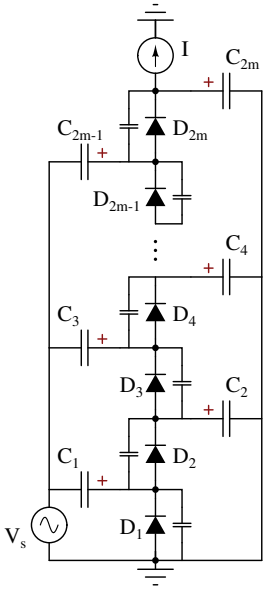


Fig. 1. Schematic of an m-stage Dickson voltage multiplier with parasitic capacitance of diodes.

only a constant parallel parasitic capacitance C_j and a constant forward voltage drop V_D as shown in the figure. The output charge per period is defined as

$$q = \frac{I}{f} \tag{1}$$

To derive a parasitic model of Dickson multiplier that can better predict the output voltage and its ripple, we first recognize that the circuit operates in a circle of 4 phases described as follows ($k = 1, ..., m$):

Phase 1: D_{2k-1} remain on, D_{2k} remain off

Phase 2: All diodes are off, transitioning to phase 3

Phase 3: D_{2k-1} remain off, D_{2k} remain on

Phase 4: All diodes are off, transitioning to phase 1

Since our focus is on the average output voltage and its ripple, it is not necessary to determine the duration of each phase. Similarly to conventional analyses, the relationships between maximum and minimum voltages across capacitors are determined by KVL in phase 1 and 3. Then averaged charge equations during phase 2 and 4 are derived to solve those voltages without complicated differential equations. We denote the voltage across capacitor C_i as V_i with the polarity defined in Fig. 1; the maximum reverse voltage blocked by diode D_i as $V_{J,i}$ (from cathode to anode) ($i = 1, ..., 2m$).

During phase 1, when the input voltage $V_{in} = -V_s$, capacitors C_{2k-1} are charged to their maximum voltages while C_{2k} (except C_{2m}) are discharged to their minimum voltages. Applying KVL to the circuit renders

$$V_{2k-1,max} = \begin{cases} V_s - V_D & k = 1 \\ V_{2k-2,min} + V_s - V_D & k \geq 2 \end{cases} \tag{2}$$

For the output capacitor C_{2m}, its voltage V_{2m} may not drop to the minimum at this point due to the load current. But at high frequency with light load, we can assume that the charge transferred between C_{2m} and the junction capacitance of D_{2m} is considerably larger than the output charge q. Hence, V_{2m} approximately reaches the minimum when $V_{in} = -V_s$. Meanwhile, diodes D_{2k} block the maximum reverse voltage that can be expressed as

$$V_{J,2k} = \begin{cases} V_{2,min} + V_D & k = 1 \\ V_{2k,min} - V_{2k-2,min} + V_D & k \geq 2 \end{cases} \tag{3}$$

Similarly during phase 2, when $V_{in} = V_s$, capacitors C_{2k} are charged to their maximum voltages while C_{2k-1} are discharged to their minimum voltages. The relationship between these voltages can be derived as

$$V_{2k,max} = V_{2k-1,min} + V_s - V_D \tag{4}$$

And the maximum reverse voltage blocked by D_{2k-1} are

$$V_{J,2k-1} = \begin{cases} V_{2,max} + V_D & k = 1 \\ V_{2k,max} - V_{2k-2,max} + V_D & k \geq 2 \end{cases} \tag{5}$$

During phases 2 and 4, capacitor voltages vary between their maximum and minimum values as V_{in} changes. For any capacitor C_i ($i = 1, ..., 2m-1$), when the voltage drops from $V_{i,max}$ to $V_{i,min}$, its charge transfer can be concluded into three categories: 1) it charges the junction capacitor of D_i from $-V_D$ to $V_{J,i}$; 2) it discharges the junction capacitor of D_{i+1} from $V_{J,i+1}$ to $-V_D$; 3) it charges C_{i+1} by q. Summarizing the charge relationship renders the averaged KCL equations

$$C_i(V_{i,max} - V_{i,min}) = C_j(V_{J,i} + V_{J,i+1} + 2V_D) + q \tag{6}$$

where C_j is the constant parasitic capacitance of diodes. During the voltage drop of C_{2m}, it only charges the junction capacitance of D_{2m} and the load. Hence, the resulting averaged charge transfer equation is

$$C_{2m}(V_{2m,max} - V_{2m,min}) = C_j(V_{J,2m} + V_D) + \frac{q}{2} \tag{7}$$

Since phases 2 and 4 are symmetrical, (6) and (7) can be derived using any of the two phases. As a result, we get $4m$ KVL equations and $2m$ averaged KCL equations from phases 1 to 4. Therefore, the unknown $4m$ capacitor voltages and $2m$ diode voltages can be solved using (2) to (7).

As a result, we can easily obtain the average output voltage and its peak-to-peak ripple by

$$\begin{aligned} V_{out} &= (V_{2m,max} + V_{2m,min})/2 \\ \Delta v_{out} &= V_{2m,max} - V_{2m,min} \end{aligned} \tag{8}$$

As predicted by (6) and (7), if we operate at high frequency with high voltage but light load conditions, voltage ripple is determined more by the ratio of junction and bypassing capacitances than the output charge. It is also worth noting that the derivation of the model does not require a sinusoidal input. As long as the input voltage has same positive and negative amplitudes, the output voltage and ripple can be calculated by (8).

To derive an input impedance of the multiplier for converter design as in [11], we further assume fundamental harmonic

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approximation for the circuit. Given a sinusoidal excitation with amplitude V_s, the multiplier can be lumped into a parallel RC circuit, where R_{eq} stands for the real power component while C_{eq} consumes the reactive power as stored in both parasitic and bypassing capacitors. Given the proposed model, the input impedance can be expressed as

$$R_{eq} = \frac{V_s^2}{I(V_{2m,max} - V_{2m,min})}$$
$$C_{eq} = \frac{1}{V_s^2} \sum_{i=1}^{2m} \left(C_i \left(\frac{\Delta V_i}{2} \right)^2 + C_j \left(\frac{V_{J,k} + V_D}{2} \right)^2 \right) \quad (9)$$

where

$$\Delta V_i = V_{i,max} - V_{i,min} \quad (10)$$

III. Description of the Proposed Power Supply

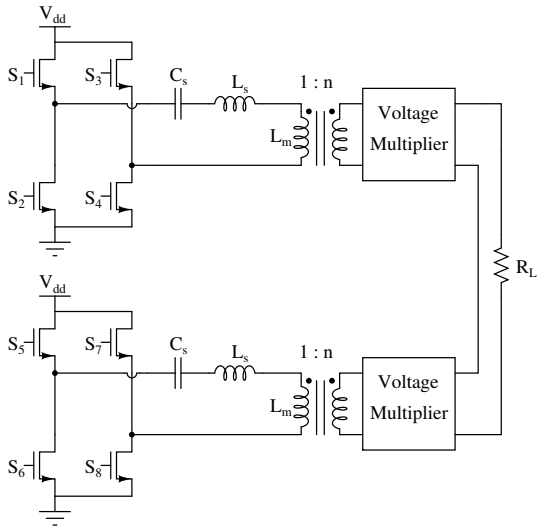


Fig. 2. Proposed implementations of the ion pump power supply. Two identical converters are connected with parallel input and series output for ripple cancellation.

The proposed ion pump power supply is shown in Fig. 2, where the load resistor R_L represents the ion pump being driven. In order to safely power an ion pump, the output voltage needs to be adjustable from 2 kV to 7 kV, while the input voltage is preferred to be around 50 V. The load can vary from its full power of 80 W to the near-vacuum condition with less than 1 mA current. As mentioned previously, the output voltage ripple at the near-vacuum condition is critical to the efficiency of the ion pump, and its peak-peak value is required to be less than 10 V.

To reduce the voltage ripple, the output of two 180° phase-shifted voltage multipliers are connected in series as shown in Fig. 2. Each voltage multiplier outputs half of the DC voltage while their ripples are shifted and cancel each other. As in Fig. 2, two 180° out-phased identical converters drive the multipliers separately with a parallel connection to the DC input.

A. Voltage Multiplier Design

Each voltage multiplier shown in Fig. 2 is implemented with the Dickson topology. Compared to Cockcroft or Hybrid multipliers, Dickson multipliers have better voltage regulation and less output ripple, especially with more stages or significant load currents. On the other hand, the Dickson topology requires higher voltage ratings for the capacitors in the circuit, where the last stage needs to withstand the full output voltage. As each voltage multiplier in the proposed design outputs a maximum voltage of 3.5 kV, four 1.5 nF, X7R, 6 kV rated ceramic capacitors are selected for the last multiplier stage, with capacitance degradation being considered. Other than that, 3 kV rated, 220 pF, C0G capacitors are used for lower voltage stages, while 4 kV rated, 1 nF, X7R capacitors are used for higher voltage stages.

As a high switching frequency is to be implemented, the diodes in the voltage multipliers should be chosen to minimize their reverse recovery loss and junction capacitance. Otherwise, the ripple reduction by increasing the frequency will be undermined, and the multipliers may suffer from lower efficiency and worse load regulation. Hence, Schottky diodes are preferred due to their very little reverse recovery loss and fast switching capabilities. However, commercially available Si Schottky diodes have no greater than 250 V voltage ratings, making them difficult to implement in high voltage circuits. Although increasing the number of multiplier stages can reduce voltage stresses across the diodes, it is at the cost of worsening the ripple and load regulation, and fewer stages are always preferred when possible. Even if the diodes are connected in series to withstand more voltage, the improved voltage rating is still limited and results in reliability issues due to unbalanced voltage distribution across the diodes [5]. On the other hand, SiC Schottky diodes have higher voltage ratings of more than 650 V, which are more suitable for high voltage applications. However, the larger junction capacitance of SiC Schottky diodes challenges the circuit design at high frequencies, and their large packages also undermine the size of the converter. With careful design considerations, SiC Schottky diode GB01SLT06-214 is selected for the proposed voltage multiplier with 8 stages, given its 650 V voltage rating, small package and small junction capacitance.

It is also worth noting that the decoupling capacitance selection in a voltage multiplier is related to the junction capacitance of diodes in the circuit, especially at light-load and high-frequency conditions. Suppose the capacitance is too small compared to the junction capacitance. In that case, the output voltage ripple will be dominated by the junction capacitance and much higher than the conventional multiplier models predicted. On the other hand, if the capacitance is selected too large, there will be a prolonged transient response in the converter, imposing difficulties in control and protection circuits for the system.

B. Full Bridge Inverter Design

As shown in Fig. 2, full-bridge inverters are implemented to convert input DC voltage V_{dd} into high frequency AC.

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Given the power level and frequency of the proposed power converter, it is of interest to use GaN transistors as the switching device due to their high-frequency capabilities and low on-resistance. To simplify the circuitry, LMG5200 80 V, 10 A GaN half-bridge power stage is selected for the proposed design, which integrates a GaN half-bridge as well as its gate driving circuits and can be directly driven by a digital signal.

C. Transformer and Resonant Tank Design

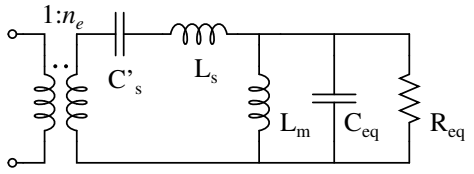


Fig. 3. The equivalent resonant tank referred to the primary side of the transformer, including a DC blocking capacitor C_s' referred to the secondary side, the cantilever inductances and an equivalent parallel RC represents the voltage multiplier.

As in Fig. 2, each transformer is represented by a cantilever model where L_s and L_m are the leakages and magnetizing inductance, and n is the effective turns ratio. Assuming fundamental harmonic approximation, Fig. 3 shows the equivalent resonant tank referred to the primary side of the transformer, including a DC blocking capacitor C_s', the cantilever inductances and an equivalent parallel RC represents the voltage multiplier.

Although represented as inductors in Fig. 3, the PCB transformer has a limited component quality factor which contributes to losses in the circuit that can be modeled as series equivalent resistance (ESR). To maximize the efficiency of the converter, it is of importance to minimize the reactive currents in the transformer windings as it results in extra losses in the ESR, especially in the primary winding. At given load conditions, i.e., output voltage and load power, the voltage and current across R_{eq} in Fig. 3 is determined. Hence, to minimize the reactive current in L_s, it is beneficial to have L_m and C_{eq} resonant at the switching frequency, which renders:

$$f_s = \frac{1}{2\pi L_m C_{eq}} \tag{11}$$

where f_s is the switching frequency. In practice, as the junction capacitance of diodes in the multiplier is nonlinear with the voltage, C_{eq} will decrease as the output voltage steps up. Additionally, if the switching frequency is beyond the resonance of L_m and C_{eq}, there will be extra capacitive current in the circuit, which is harmful to the zero-voltage-switching (ZVS) conditions for full-bridge transistors and may result in significant losses at high frequencies. Therefore, f_s should be chosen close to but less than the resonance of L_m and C_{eq}.

The DC blocking capacitor C_s', as well as the leakage inductor L_s, act as a filter to suppress 3rd and higher harmonics in the circuit. While tuning the resonance of L_s and C_s' to be precisely the switching frequency that gives the highest voltage gain, it is desired to have enough inductive current to maintain ZVS conditions for the transistors, and thus f_s should be chosen higher than the resonance of L_s and C_s'. For convenience, it is also possible to have a large C_s' that blocks DC current but bypasses all the high-frequency components. It is also worth noting that, in order to maintain a valid fundamental harmonic approximation, the series resonant tank L_s and C_s' should display a high enough impedance at 3rd and higher harmonics. Otherwise, the power converter may suffer significant losses from the harmonics and achieve poor efficiency. As an extreme example, if the resonance of $L_m \| L_s$ and C_{eq} is right at $3f_s$, there will be a 3rd harmonic short in the circuit and may damage the converter. If the leakage inductance is too small due to the tight coupling between windings, an extra inductor can be added to the circuit for harmonic suppressing at the cost of reducing the voltage gain.

Ideally, assuming the resonance of L_s and C_s' at switching frequency, the voltage gain of the power converter will be

$$V_{out} = 2mn\frac{4}{\pi}V_{dd} \tag{12}$$

However, this is a too optimistic approximation, and three major mechanisms undermine the maximum voltage gain. Except for the aforementioned resonance deviations in the resonant tank and the nonlinear C_{eq}, the voltage multiplier suffer from significant load regulation issues at high frequency, which can degrade the voltage gain by 20%. Hence, a more accurate model for voltage multipliers must be included to determine the voltage gain and the self inductance of the secondary winding.

Given the diodes and capacitors selected in section III.A, especially the junction capacitance of SiC Schottky diodes, the voltage gain of the proposed multiplier can be determined. According to the model of voltage multipliers, the voltage gain is around 12.5 when the output is 3.5 kV with 1 mA load and a switching frequency between 1 to 2 MHz. Similarly, the equivalent capacitance is $C_{eq} \approx 200$ pF. Hence, assuming the input voltage to be 50 V, the equivalent turn ratio should be

$$n_e = \frac{\pi V_{out}}{12.5 \times 4V_{dd}} = 4.4 \tag{13}$$

To have a reasonable inductance for designing transformer geometries, the switching frequency is chosen to be 1.35 MHz, and the resonant frequency for L_m and C_{eq} is slightly higher and determined to be 1.85 MHz (500 kHz higher). As a result, assuming $k = 0.85$ for a 1.6 mm PCB transformer, the self inductance of the primary winding is approximately

$$L_1 = \frac{1}{4\pi^2 f^2 k^2 n_e^2 C_{eq}} \approx 2.65\mu H \tag{14}$$

As a result, the self inductance of the primary winding is approximately

$$L_2 = n_e^2 k^2 L_1 \approx 37\mu H \tag{15}$$

In practice, the transformer in Fig. 2 can be implemented on a four-layer PCB board with the primary winding at the top and the secondary winding at the bottom. The inner layers can be used for connecting the terminals.

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To take advantage of all the board areas, it is of interest to implement the PCB transformers with a square shape. Mohan et al. proposed an analytical method for calculating planar spiral inductances with given geometries, and for a square-shape inductor, its inductance can be expressed as [10]:

$$L = 0.635\mu_0 N^2 D_{avg}\left[\ln\left(\frac{2.07}{\rho}\right) + 0.18\rho + 0.13\rho^2\right] \quad (16)$$

where N is the number of turns, $D_{avg} = (D_{out} + D_{in})/2$ is the average of outer and inner width and $\rho = (D_{out} - D_{in})/(D_{out} + D_{in})$ is the geometry factor. Although this formula gives a close approximation for self inductances of transformer windings, the coupling coefficient cannot be determined. In general, it is more challenging to develop accurate and simple analytical solutions for the coupling coefficient, and numerical methods are preferred. As the proposed transformer is designed to be air-core, we use FastHenry simulation tool to determine the coupling between the windings [12].

The outer width of the windings is chosen to be 6 cm, which fits the geometry limits for the power controller. The primary trace width is 1.524 mm, while the secondary trace width is set to 0.25 mm. Since the skin depth of copper is 56 μm, using 2 oz (70 μm) should result in higher quality factors in windings. According to the simulation, with the geometry factor $\rho = 0.6$ and given 1.6 mm PCB thickness, the inductance matrix that approximately matches the design is:

$$L = \begin{bmatrix} 2.6\mu H & 8\mu H \\ 8\mu H & 35.8\mu H \end{bmatrix} \quad (17)$$

The resulting coupling coefficient is 0.83, which is close to our assumption of 0.85 for 1.6 mm boards. Besides the inductance, FastHenry also gives the quality factor of coils by taking high-frequency loss effects into consideration. As a result, the transformer has a quality factor $Q = 75$ for its primary, while $Q = 61$ for its secondary. All the quality factors are calculated at the switching frequency of 1.35 MHz.

IV. Experimental Verification

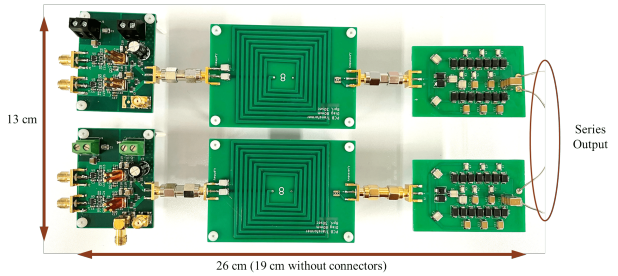


Fig. 4. Photograph of the proposed ion pump power supply prototype

Fig. 4 shows the photograph of the proposed power supply prototype, of which the load is a resistor representing the ion pump. The converter operates at 1.35 MHz with a 20-60 V supply voltage. Key components implemented in the prototype are listed in Tab. I. The experimentally measured inductance matrix of transformer is

$$L = \begin{bmatrix} 2.77\mu H & 8.9\mu H \\ 8.9\mu H & 39.16\mu H \end{bmatrix} \quad (18)$$

and the quality factors measured at 1.35 MHz are 64 for the primary and 48 for the secondary, which matches the results predicted by FastHenry simulations.

For full load (80 W load) ripple measurement, the load resistance is set to 600 kohm at 7 kV and 50 kohm at 2 kV. Tab. II summarize the experimental results. The peak efficiency is approximately 84% when the output is 7 kV with gating power included.

For light load (1 mA load) ripple measurement, the load resistance is set to 7 Mohm at 7 kV and 2 Mohm at 2 kV. The resulting ripples shown in Fig. 5 and 6 are measured with CalTest CT4028 high voltage probe under AC coupling. The proposed prototype achieve approximately 3 Vpp ripple at 2 kV (5 Vpp including spikes due to parasitic inductances at probe) and 9 Vpp at 7 kV. Compared to the design proposed by Bae et. al. operating at similar voltages and frequencies, the proposed design has around 10 times smaller ripple of 0.1% without increasing the converter size [4].

TABLE I
Key Components of the Proposed Converter

Device Symbols	Component Description
S_{1-8}	LMG5200 80 V 10 A GaN Power Stage
D_{1-16}	GB01SLT06-214 650 V 1 A SiC Schottky Diodes
C_{1-6}	220 pF C0G, Multiplier Capacitor
C_{7-10}	200 pF C0G, Multiplier Capacitor
C_{11-15}	1 nF X7R, Multiplier Capacitor
C_{16}	6 nF X7R, Multiplier Capacitor
C_s	25 nF C0G, DC-Blocking Capacitor

TABLE II
Experimental Results

V_{dd}	I_{dd}	V_{out}	R_L	P_{gate}	η
60.4 V	1.62 A	7.0 kV	597.9 kΩ	0.22 W	83.6 %
26.0 V	4.02 A	2.0 kV	49.9 kΩ	0.22 W	76.5 %

V. Conclusion

In this paper, we propose a detailed design of ion pump power supply to achieve high efficiency while maintaining

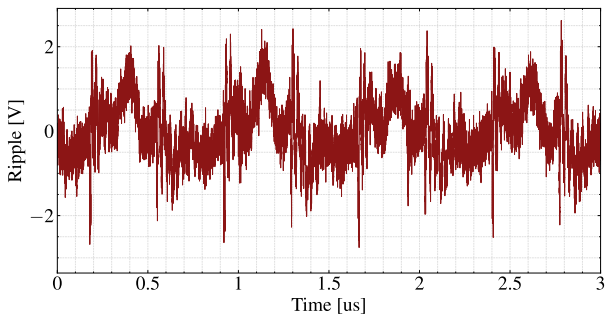


Fig. 5. Output Voltage ripple measured at 2 kV with 1 mA light load

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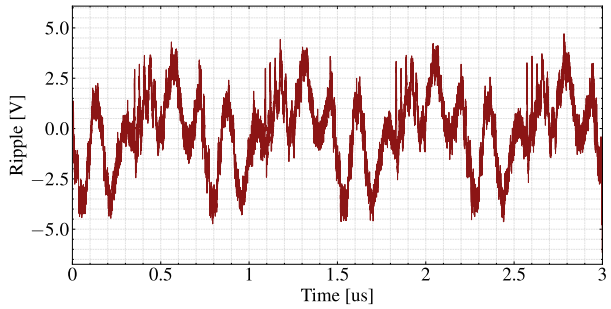


Fig. 6. Output Voltage ripple measured at 7 kV with 1 mA light load

low output ripple. The proposed prototype consists of two identical converters with 180° phase shift for ripple cancelling. The converters are implemented with PCB transformers and Dickson voltage multipliers. A detailed model for Dickson voltage multiplier is also proposed in this paper to simplify the design. The prototype is experimentally implemented and tested with 1.35 MHz switching frequency. The maximum output power is 80 W with 84 % peak efficiency, and the ripple is approximately 0.1% when the output current is less than 1 mA.

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